

Software Test Plan

Quality Assurance & Verification Strategy for SheGuidesMe Platform

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Status: Approved / Ready for Execution

Application: Web, iOS & Android

Executive Summary

This Master Test Plan establishes the complete quality assurance strategy, execution framework, test environments, criteria, and governance required to validate the **SheGuidesMe** ecosystem. The primary objective is to deliver a secure, highly accessible, resilient, and trust-centered platform for female travelers and local guides.

1. Introduction & Objectives

1.1 Purpose

The purpose of this Test Plan is to define the exact testing strategy, scope, resources, timeline, and deliverables for verifying and validating the **SheGuidesMe** web and mobile applications. It serves as the single source of truth for QA engineers, developers, product managers, and security auditors.

1.2 Business Core & System Overview

SheGuidesMe is a multi-tier platform built to empower women travelers by connecting them with verified local female guides and community hosts. The system architecture comprises four major layers:

- Traveler Client (Web & Native Mobile - iOS/Android):** Features location discovery, interactive map, real-time booking, chat, and emergency SOS alerts.
- Guide Portal & App:** Identity verification submission, availability calendar management, booking request response, and earnings/payout analytics.
- Admin Back-Office System:** Identity verification queue moderation, user dispute resolution, emergency alert monitoring, refund processing, and audit logging.
- Integration Ecosystem:** Third-party integrations for Identity Verification (Persona/Onfido), Payment & Escrow Processing (Stripe Connect), Maps & Geolocation (Google Maps API), and Communications (Twilio / SendGrid).

2. Scope of Testing

2.1 In-Scope Functional Modules

Module Area	Functional Coverage	Target Platforms
Authentication & Onboarding	Sign-up/In via Email, Social SSO (Google, Apple), MFA, Password Reset, Role Switch.	Web, iOS, Android
Identity Verification (Trust Engine)	Government ID upload, liveness/selfie check, background check webhook processing, verified badge assignment.	Web, iOS, Android, Admin
Search & Discovery	Geo-location search, interactive radius map, multi-attribute filtering (languages, activities, price, ratings).	Web, iOS, Android

Module Area	Functional Coverage	Target Platforms
Booking Engine	Slot reservation, instant booking, approval workflows, timezone offset conversions, auto-cancellation timers.	Web, iOS, Android
Payments & Escrow	Pre-authorization, escrow holds, payout disbursements, platform commission deductions, multi-currency conversion.	Web, Mobile, Stripe Sandbox
In-App Safety & Messaging	Real-time messaging, PII masking (blocking phone numbers/emails pre-booking), location sharing, Emergency SOS broadcast.	iOS, Android
Reviews & Community	Post-trip review creation (verified bookings only), rating calculation, abuse reporting, content moderation.	Web, iOS, Android, Admin

2.2 In-Scope Non-Functional Areas

- **Security & Privacy:** OWASP Top 10 vulnerabilities, BOLA/IDOR checks, TLS 1.3 verification, GDPR/CCPA PII data isolation.
- **Performance & Scalability:** Load testing up to 10,000 concurrent user sessions; API latency benchmarking (<300ms SLA).
- **Compatibility & Responsiveness:** Mobile OS (iOS 15+, Android 11+), Desktop Browsers (Chrome, Safari, Firefox, Edge).
- **Accessibility:** WCAG 2.1 AA screen reader compatibility and visual contrast standard checks.

2.3 Out-of-Scope

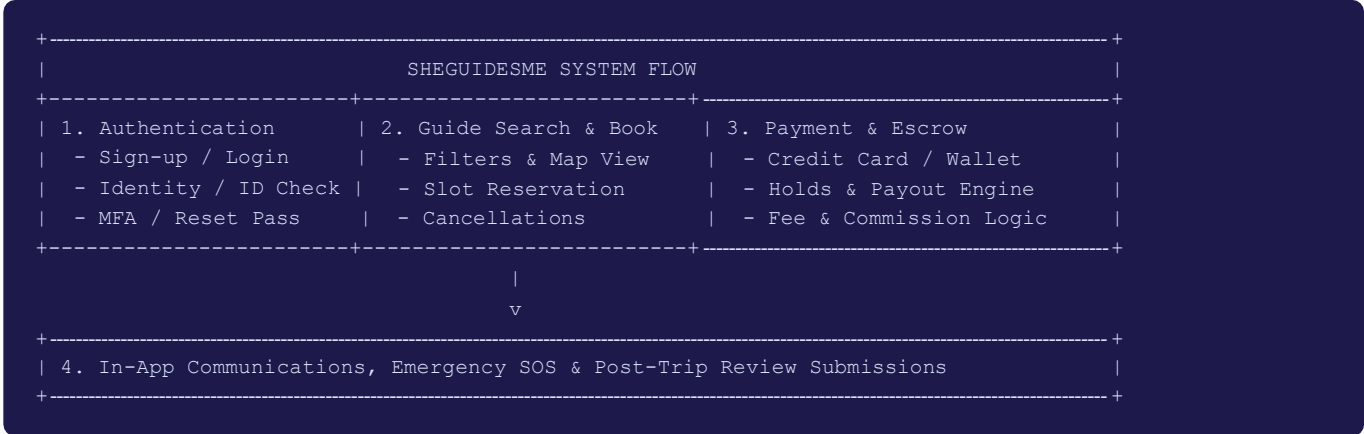
- Internal testing of vendor backend hardware/networks (e.g., Stripe's internal payment routing logic).
- Legacy web browsers (Internet Explorer, Safari v13 or below).
- Third-party sandbox infrastructure load testing.

3. Test Strategy & Execution Methodology

3.1 Testing Levels & Responsibilities

Level	Primary Objectives	Owner	Tooling
Unit Testing	Validate core business logic functions, mathematical calculations, and utility helpers.	Engineering Team	PyTest
Integration Testing	Verify API endpoints, database schemas, and external webhook listener handlers.	QA & Backend Team	Postman
End-to-End (E2E)	Automate complete user journeys from registration to payment and review submission.	QA Automation Team	Selenium, Playwright
Security Testing	Uncover authentication bypasses, sensitive data leaks, and authorization flaws.	Security Lead	OWASP ZAP, Burp Suite
Load & Stress	Measure system response under traffic spikes and locate bottleneck thresholds.	DevOps / QA	JMeter
UAT	Verify business alignment and real-world usability with internal stakeholders.	Product Managers	Exploratory Scripts

3.2 System Architecture Test Coverage Map



4. Test Deliverables & Milestones

The QA team will produce the following artifacts throughout the software development lifecycle:

- 1. **Master Test Plan Document** (This artifact)
- 2. **Test Case Repository & Traceability Matrix** (Maintained in Zephyr / TestRail)
- 3. **Automated Test Execution Suites** (Playwright & Appium repository scripts)
- 4. **Defect Reports** (Logged and tracked in Jira)
- 5. **Performance & Vulnerability Assessment Reports**
- 6. **Final Quality Sign-Off & Release Recommendation Summary**

5. Environment Requirements & Infrastructure

5.1 Test Environment Architecture

Testing will occur across three isolated environments before production deployment:

- **Staging / QA:** Continuous deployment target. Connected to mock APIs for ID verification and sandbox payment gateways. Seeded with anonymized test datasets.
- **UAT / Pre-Prod:** Mirror copy of production architecture. Uses live third-party vendor sandbox environments (Stripe Test Keys, Persona Test Engine).
- **Production:** Automated sanity smoke suite runs immediately post-deployment using dedicated test canary accounts.

5.2 Tooling Stack

Category	Tool Standard	Configuration / Details
Test Management	Zephyr Scale / Jira	Integrated with GitHub PR checks
Web Automation	Playwright (TypeScript)	Cross-browser parallel execution
Mobile Automation	Appium + WebdriverIO	Execution on BrowserStack Cloud Device Farm
API & Performance	Postman, Newman, k6	Automated API test runs in CI/CD pipeline
Defect Tracking	Jira Software	Automated defect generation from test failures

6. Key Functional Test Cases & Traceability Matrix

Test ID	Module	Test Objective	Expected Outcome	Priority
TC-AUTH-001	Auth	Verify user registration with valid credentials & email confirmation link.	User receives verification email; link activates account successfully.	P0 - CRITICAL
TC-AUTH-002	Auth/Trust	Verify Guide profile remains hidden in search until ID verification is approved.	Guide profile state is 'Pending Verification' and invisible on public search.	P0 - CRITICAL
TC-SRCH-001	Search	Verify radius search correctly filters guides within 15km of target city.	Only verified guides with matching locations are rendered on map and list.	P1 - HIGH
TC-BOOK-001	Booking	Verify double-booking prevention for overlapping calendar time slots.	Slot shows unavailable; booking API rejects simultaneous request with 409 Conflict.	P0 - CRITICAL
TC-PAY-001	Payments	Verify pre-authorization payment hold upon booking request placement.	Payment status in Stripe shows 'Requires Capture'; funds reserved in escrow.	P0 - CRITICAL
TC-SAFE-001	Safety	Verify auto-masking of phone numbers and external links in chat pre-booking.	Sensitive text patterns replaced with [Redacted for Safety] notice.	P1 - HIGH
TC-SAFE-002	Safety	Verify Emergency SOS button triggers immediate location notification to contacts.	SMS and push notification dispatched with precise GPS lat/long link within 5 sec.	P0 - CRITICAL

7. Entry, Exit & Suspension Criteria

7.1 Entry Criteria

1. Feature code build successfully compiled and deployed to Staging.
2. All unit tests pass with a minimum threshold of 80% code coverage.
3. Test data (guide profiles, traveler accounts, active sandbox tokens) successfully seeded.

7.2 Suspension & Resumption Criteria

- **Suspension:** Testing stops if a **P0 (Critical)** blocker prevents access to primary test workflows (e.g., authentication system failure, continuous database crash) and no workaround exists.
- **Resumption:** Testing resumes once a fix is deployed, verified via smoke testing, and environment stability is confirmed.

7.3 Exit & Release Criteria

1. 100% execution of all planned high-priority functional and non-functional test cases.
2. Minimum overall test suite pass rate of 95%.
3. **Zero open P0 (Critical) or P1 (High) defects.**
4. All P2 (Medium) and P3 (Low) open issues fully documented, categorized, and formally accepted by Product Management for patch releases.
5. Formal sign-off provided by QA Lead, Engineering Manager, and Product Manager.

8. Defect Management & Classification Framework

Defects identified during testing will be logged in Jira following a standardized template (Summary, Steps to Reproduce, Expected vs. Actual Result, Screenshots, Network Logs, System Info). Defects are classified according to the following SLA matrix:

Severity Level	Description & Impact	Target SLA	Example Scenario
P0 - CRITICAL	System unavailability, data breach risk, core transaction flow blocked.	< 2 Hours	Payment gateway crashes; SOS button fails to trigger SMS alerts.
P1 - HIGH	Major feature broken with no feasible workaround available.	< 24 Hours	Guide cannot update calendar availability; ID document upload fails on iOS.
P2 - MEDIUM	Non-critical feature issue; workaround is available to the user.	< 3 Days	Filter dropdown visual misalignment; review sorting order error.
P3 - LOW	Cosmetic errors, minor visual glitches, copy typos.	Next Sprint	Spelling typo in FAQ footer; minor color contrast discrepancy.

9. Approval & Quality Sign-Off

By signing below, the undersigned stakeholders accept this Software Test Plan and agree that adherence to the criteria outlined herein governs the production release decision for the SheGuidesMe platform.

Role	Name & Title	Signature / Status	Date
QA Lead	Quality Assurance Manager	APPROVED	September 5, 2026
Engineering Lead	VP of Engineering	APPROVED	September 5, 2026
Product Owner	Head of Product	APPROVED	September 5, 2026
Trust & Safety Lead	Director of Operations	APPROVED	September 5, 2026